

## 1234 M001 Week 3 — English Worksheet

**Topic:** Variables · **Course:** Pyramid Problems · **Time:** about 45 minutes

### 1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
set size to 5
repeat size times:
  place block down
  move forward
```

**How many blocks does the agent place?**

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|  |

```
set size to 3
repeat size times:
  place block down
  move forward
set size to 6
repeat size times:
  place block down
  move forward
```

**How long is the full line of blocks? Why?**

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```
set height to 4
repeat height times:
  place block down
  move up by 1
```

**What does the agent build? How tall?**

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## 2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

**Bug A** — The agent is supposed to build a row of 7 blocks using a variable called `size`.

```
repeat 7 times:
  place block down
  move forward
```

Write the fixed code using a variable:

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Why is using a variable better here?

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**Bug B** — The agent is supposed to build a row of `size` blocks. The size should be 5.

```
set size to 5
repeat 3 times:
  place block down
  move forward
```

Write the fixed code:

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**Why was it wrong? Why does your fix work?**

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**Bug C** — The agent is supposed to build a tower that is `height` blocks tall, where `height` is 6.

```
set height to 6
repeat height times:
  place block down
  move forward
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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### 3 · Tell Me What You Built 📷

Now switch to your homework world. Use a `size` variable to build a **size-5 wall**, then change the variable to **12** and run it again. Compare the two results. When you finish, come back here.

Send a photo or video of what you built, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

I made a variable called ...

I set it to ...

I used the variable in my loop because ...

The hardest part was ...

If I changed the variable to a bigger number, the build would ...

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## 4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you walk the camera around what you built. Talk through it like you are teaching someone who has never seen it. Try to use these words: **variable, set, loop, repeat, block.**

1. Show what you built.
2. Read your variable and loop out loud and say what each one does.
3. Change the variable to a different number and run the code again — show the new result.
4. Say why variables are useful.

**Write what you will say in your video.** Use the space below to plan it before you record — you can read from it while filming.

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**Submit** 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.